

Human-readable derivation of the degree-3 tensor identity

Repository source: `proof/DEGREE3_TENSOR_IDENTITY.md`

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Status

This file supplies a hand-auditable derivation of the central algebraic lemma used in `proof/BRIDGE_LEMMAS.md`. The derivation is independent of the final volume arithmetic. Exact symbolic checks remain in `python/verify_degree3_identity.py` and `python/verify_degree3_identity_alt.py`; the purpose here is to explain why the identities are true rather than merely report a computer expansion.

All indices range over $\{1,2,3\}$ and repeated indices are summed. The metric is the Euclidean metric, so raising and lowering indices has no effect.

1. Statement of the degree-3 lemma

Let $\mathbf{T}$ be a real symmetric trace-free rank-3 tensor on $\mathbb{R}^3$:

$$T_{ijk} = T_{(ijk)}, \quad T_{iik} = 0. \quad (1.1)$$

Define the homogeneous cubic

$$F(x) := T_{ijk} x_i x_j x_k \quad (1.2)$$

and let

$$Y := F|_{S^2}.$$

For $\mathbf{u}$ in S^2 , set

$$C(u) := A_Y(u) = \nabla_{S^2}^2 Y(u) + Y(u) I_{T_u S^2}. \quad (1.3)$$

Introduce the tensor invariants

$$\tau := T_{ijk}T_{ijk}, \quad B_{ij} := T_{ikl}T_{jkl}, \quad B_0 := B - \frac{\tau}{3}I. \quad (1.4)$$

Since $\text{tr } B = \tau$, the matrix B_0 is traceless.

The goal is to prove

$$\int_{S^2} |C|^2 d\sigma = \frac{176\pi}{7} \tau \quad (1.5)$$

and, with the determinant taken on the two-dimensional tangent plane,

$$\int_{S^2} (4 \det C)^2 d\sigma = \frac{\pi}{1001} (555264 \tau^2 - 2265600 |B_0|^2). \quad (1.6)$$

It follows immediately that

$$\|4 \det C\|_{L^2(S^2)}^2 \leq \frac{15183}{17303\pi} \|C\|_{L^2(S^2)}^4. \quad (1.7)$$

2. Harmonic cubics and symmetric trace-free tensors

Because T is symmetric,

$$\partial_a F = 3T_{ajk}x_jx_k, \quad \partial_a \partial_b F = 6T_{abk}x_k. \quad (2.1)$$

Consequently,

$$\Delta_{\mathbb{R}^3} F = 6T_{iik}x_k. \quad (2.2)$$

Thus F is harmonic if and only if T is trace-free. Every degree-3 spherical harmonic is the restriction of a unique harmonic homogeneous cubic, so (1.2) represents an arbitrary element of $\mathbb{H}_3$.

3. The spherical curvature tensor

Fix u in S^2 and write

$$P := I - u \otimes u, \quad (3.1)$$

the orthogonal projection onto $T_u S^2 = u^\perp$. Define

$$M(u)_{jk} := T_{ijk}u_i, \quad v(u)_k := T_{ijk}u_iu_j = M(u)u, \quad Y(u) = u^T M(u)u. \quad (3.2)$$

The Euclidean Hessian of F at u is

$$D^2 F(u) = 6M(u). \quad (3.3)$$

Lemma 3.1 (restriction Hessian formula)

For tangent vectors ξ, η in $T_u S^2$,

$$\nabla_{S^2}^2 Y(\xi, \eta) = D^2 F(u)[\xi, \eta] - 3Y(u)\langle \xi, \eta \rangle. \quad (3.4)$$

Proof Let $\mathbf{x}$ and $\mathbf{z}$ be tangent vector fields with $\mathbf{x}(u)=\mathbf{x}_i$ and $\mathbf{z}(u)=\mathbf{e}_a$. For the unit sphere,

$$D_X Z = \nabla_X^{S^2} Z - \langle X, Z \rangle u. \quad (3.5)$$

Therefore

$$\begin{aligned} \nabla_{S^2}^2 Y(\xi, \eta) &= X(ZF) - (\nabla_X^{S^2} Z)F \\ &= D^2 F(u)[\xi, \eta] + DF(u)(D_X Z - \nabla_X^{S^2} Z) \\ &= D^2 F(u)[\xi, \eta] - \langle \xi, \eta \rangle DF(u)[u]. \end{aligned}$$

Euler's identity for the homogeneous cubic gives $DF(u)[u]=3F(u)=3Y(u)$, which proves (3.4). $\square$

Adding $Y \cdot \mathbf{I}$ to (3.4) gives the tangent endomorphism

$$\boxed{C = 6PMP - 2YP.} \quad (3.6)$$

Here and below we regard a tangent endomorphism as a 3×3 symmetric matrix that vanishes on the normal direction u . Its nonzero eigenvalues are exactly the two tangent eigenvalues.

4. A compact formula for $4 \det C$

Trace-freeness of T implies

$$\text{tr } M = T_{ijj} u_i = 0. \quad (4.1)$$

Since $P=I-u \cdot u^T$,

$$\text{tr}(PMP) = \text{tr}(MP) = \text{tr } M - u^T M u = -Y. \quad (4.2)$$

Choose an orthonormal basis whose first vector is u . In block form,

$$M = \begin{pmatrix} Y & w^T \\ w & N \end{pmatrix}, \quad v = M u = \begin{pmatrix} Y \\ w \end{pmatrix}, \quad PMP = \begin{pmatrix} 0 & 0 \\ 0 & N \end{pmatrix}.$$

It follows that

$$|PMP|^2 = |M|^2 - 2|v|^2 + Y^2. \quad (4.3)$$

Using (3.6), (4.2), $|P|^2 = \text{tr } P = 2$, and the fact that PMP is tangent,

$$\text{tr } C = 6 \text{tr}(PMP) - 2Y \text{tr } P = -10Y, \quad (4.4)$$

$$\begin{aligned} |C|^2 &= 36|PMP|^2 - 24Y \langle PMP, P \rangle + 4Y^2 |P|^2 \\ &= 36|PMP|^2 - 24Y \text{tr}(PMP) + 8Y^2 \\ &= 36|M|^2 - 72|v|^2 + 68Y^2. \end{aligned} \quad (4.5)$$

If the tangent eigenvalues of C are μ_1, μ_2 , then the 3×3 extension has eigenvalues $\mu_1, \mu_2, 0$. Hence

$$4 \det C = 2((\text{tr } C)^2 - \text{tr}(C^2)). \quad (4.6)$$

Substitution of (4.4)-(4.5) yields

$$\boxed{q(u) := 4 \det C(u) = 64Y(u)^2 + 144|v(u)|^2 - 72|M(u)|^2.} \quad (4.7)$$

This is the compact determinant formula checked independently by the two exact Python scripts and by the Python and R numerical tests.

5. Spherical moments

For $2m$ indices in dimension three,

$$\int_{S^2} u_{i_1} \cdots u_{i_{2m}} d\sigma = \frac{4\pi}{(2m+1)!!} \sum_{\mathcal{P}} \prod_{\{a,b\} \in \mathcal{P}} \delta_{i_a i_b}, \quad (5.1)$$

where the sum runs over all pairings $\mathcal{P}$ of $\{1, \dots, 2m\}$. Odd moments vanish. To see (5.1), let $\mathbf{G} = \mathbf{R}\mathbf{U}$ be a standard Gaussian vector in $\mathbb{R}^3$, with $\mathbf{U}$ uniform on S^2 and independent of the radius $\mathbf{R}$. Wick's Gaussian moment formula gives

$$\mathbb{E}[G_{i_1} \cdots G_{i_{2m}}] = \sum_{\mathcal{P}} \prod_{\{a,b\} \in \mathcal{P}} \delta_{i_a i_b},$$

while $\mathbb{E}[\mathbf{R}^{2m}] = (2m+1)!!$ and

$$\mathbb{E}[U_{i_1} \cdots U_{i_{2m}}] = \frac{1}{4\pi} \int_{S^2} u_{i_1} \cdots u_{i_{2m}} d\sigma.$$

Dividing the Gaussian identity by $\mathbb{E}[\mathbf{R}^{2m}]$ proves (5.1).

Three low-order consequences will be used first. Since a trace formed inside a single copy of $\mathbf{T}$ vanishes,

$$\int_{S^2} Y^2 d\sigma = \frac{4\pi}{7!!} (3!) \tau = \frac{8\pi}{35} \tau, \quad (5.2)$$

$$\int_{S^2} |v|^2 d\sigma = \frac{4\pi}{5!!} (2) \tau = \frac{8\pi}{15} \tau, \quad (5.3)$$

$$\int_{S^2} |M|^2 d\sigma = \frac{4\pi}{3} \tau. \quad (5.4)$$

Indeed, in (5.2) all three indices of one copy of $\mathbf{T}$ must be paired with the three indices of the other, giving $3!$ pairings. In (5.3), the two nonzero pairings cross between the two copies of $\mathbf{T}$. Equation (5.4) follows from $|\mathbf{M}|^2 = \mathbf{B}_{ij} u_i u_j$ and $\text{tr } \mathbf{B} = \tau$.

Now integrate (4.5):

$$\begin{aligned} \int_{S^2} |C|^2 d\sigma &= 36 \frac{4\pi}{3} \tau - 72 \frac{8\pi}{15} \tau + 68 \frac{8\pi}{35} \tau \\ &= \frac{176\pi}{7} \tau. \end{aligned} \quad (5.5)$$

This proves (1.5).

6. The two quartic contractions

Besides τ , introduce

$$J := \text{tr}(B^2) = B_{ij} B_{ij}. \quad (6.1)$$

and the tetrahedral contraction

$$K := T_{abc} T_{ade} T_{bdf} T_{cef}. \quad (6.2)$$

The moment expansion of a product of four copies of $\mathbf{T}$ naturally produces three contraction graphs. Representative index contractions are

$$D = (T_{abc}T_{abc})(T_{def}T_{def}) = \tau^2,$$

$$J = T_{abc}T_{dbc}T_{aef}T_{def} = \text{tr}(B^2),$$

and the complete tetrahedral contraction $\mathbb{K}$ in (6.2). By symmetry of $\mathbb{T}$ and relabeling of dummy indices, every graph of the same type has the same value. In dimension three the three values are related.

Lemma 6.1 (dimension-three quartic relation)

$$\boxed{K = \frac{1}{2}\tau^2 - J.} \quad (6.3)$$

Proof For each i , let $\mathbf{A}_i$ be the symmetric traceless 3×3 matrix

$$(A_i)_{jk} := T_{ijk}. \quad (6.4)$$

Symmetry of $\mathbf{A}_i$ follows from symmetry of $\mathbb{T}$, and $\text{tr } \mathbf{A}_i = \mathbb{T}_{\{i j j\}} = 0$ follows from (1.1).

Then

$$\tau = \sum_i \text{tr}(A_i^2), \quad J = \sum_{i,j} (\text{tr}(A_i A_j))^2, \quad (6.5)$$

and symmetry of $\mathbb{T}$ gives

$$\sum_i A_i^2 = B. \quad (6.6)$$

To verify (6.6), compare components:

$$\left(\sum_i A_i^2 \right)_{jk} = T_{ijl} T_{ilk} = T_{jil} T_{kil} = B_{jk}.$$

For every traceless 3×3 matrix $\mathbf{X}$, Cayley-Hamilton reads

$$X^3 - \frac{1}{2} \text{tr}(X^2)X - \frac{1}{3} \text{tr}(X^3)I = 0.$$

Multiplying by $\mathbf{X}$ and taking the trace gives

$$\text{tr}(X^4) = \frac{1}{2} (\text{tr}(X^2))^2. \quad (6.7)$$

Apply (6.7) to $\mathbf{X} = \mathbf{A} + \mathbf{tD}$ and compare the coefficient of $\mathbf{t}^2$. Cyclicity of trace gives

$$4 \text{tr}(A^2 D^2) + 2 \text{tr}(ADAD) = \text{tr}(A^2) \text{tr}(D^2) + 2 (\text{tr}(AD))^2. \quad (6.8)$$

Set $\mathbf{A} = \mathbf{A}_i$, $\mathbf{D} = \mathbf{A}_j$, and sum over i, j . The left-hand tetrahedral term is

$$\sum_{i,j} \text{tr}(A_i A_j A_i A_j) = K, \quad (6.9)$$

where equality follows by writing out the indices and using the full symmetry of $\mathbb{T}$. Also, by (6.6),

$$\sum_{i,j} \text{tr}(A_i^2 A_j^2) = \text{tr} \left(\left(\sum_i A_i^2 \right)^2 \right) = \text{tr}(B^2) = J. \quad (6.10)$$

Summing (6.8) therefore gives

$$4J + 2K = \tau^2 + 2J,$$

which rearranges to (6.3). $\square$

7. Pairing count for the six quartic moments

Write

$$a := Y^2, \quad b := |v|^2, \quad c := |M|^2. \quad (7.1)$$

We need the six integrals of $a^2, ab, ac, b^2, bc, c^2$.

Represent the four copies of $\mathbf{T}$ by four labeled vertices. Each tensor has three index slots, so every complete contraction is a 3-regular multigraph on those four vertices. An index contraction within one tensor is a loop and vanishes because $\mathbf{T}$ is trace-free. Every surviving loopless graph is one of:

- $\mathbf{D}$: two disjoint pairs joined by three parallel edges, contributing τ^2 ;
- $\mathbf{J}$: two opposite doubled edges and two connecting single edges, contributing J ;
- $\mathbf{K}$: the simple complete graph on four vertices, contributing K .

There are no other possibilities. A triple edge forces the disconnected type. If a connected graph has no multiple edge, it is $\mathbf{K}_4$. If it has a double edge, degree three forces a second opposite double edge and two single connecting edges.

Some contractions are already present before applying the spherical moment formula. In the table, 12 denotes one fixed contraction between tensor vertices 1 and 2, and 12^2 denotes two such fixed contractions. The remaining $\mathbf{u}$ -indices are paired by (5.1).

integrand	number 2m of u-indices	fixed edges	n _D	n _J	n _K	after using K= $\tau^2/2$ -J: (n _{tau} , n _J)
Y^4	12	none	108	1944	1296	(756, 648)
$Y^2 v ^2$	10	34	12	216	144	(84, 72)
$Y^2 M ^2$	8	34^2	6	36	0	(6, 36)
$ v ^4$	8	12, 34	4	40	16	(12, 24)
$ v ^2 M $	6	12, 34^2	2	8	0	(2, 8)
$ ^2$						
$ M ^4$	4	$12^2, 34^2$	1	2	0	(1, 2)

Here

$$n_\tau = n_D + \frac{1}{2}n_K, \quad n'_J = n_J - n_K. \quad (7.2)$$

For completeness, the counts can be obtained directly as follows.

1. Y^4 . The disconnected type has $3(3!)^2=108$ pairings. The tetrahedral type has $(3!)^4=1296$. There are six labeled $\mathbf{J}$ graphs; each has $3^4(2!)^2=324$ slot assignments, giving 1944.
2. $Y^2|v|^2$. The fixed edge is 34. Type $\mathbf{D}$ gives $3!2!=12$, and type $\mathbf{K}$ gives $3!3!2!2!=144$. For type $\mathbf{J}$, two graphs use 34 as a doubled edge and contribute 72 each; two use it as a single edge and contribute 36 each. Total: 216.
3. $Y^2|M|^2$. The two fixed 34 edges force either type $\mathbf{D}$, with $3!=6$, or one of two $\mathbf{J}$ graphs, each contributing $3*3*2=18$.

4. $|v|^4$. Fixed edges are 12 and 34. Type D contributes $(2!)^2=4$; type K contributes $(2!)^4=16$. Two J graphs use the fixed matching as doubled edges and contribute 16 each; two use it as single edges and contribute 4 each. Total: 40.
5. $|v|^2|M|^2$. Fixed edges are 12 and 34^2 . Type D contributes $2!=2$; the two possible J graphs contribute 4 each.
6. $|M|^4$. Fixed edges are 12^2 and 34^2 . The three pairings of the four remaining slots give one D contraction and two J contractions.

The script `python/verify_pairing_counts.py` enumerates all pairings and checks this table exactly; `R/verify_pairing_counts.R` is the matching base-R check.

Multiplying the reduced counts by the factor $4\pi/(2m+1)!!$ from (5.1) gives

$$\begin{aligned}
\int Y^4 d\sigma &= \pi \left(\frac{16}{715} \tau^2 + \frac{96}{5005} J \right), \\
\int Y^2 |v|^2 d\sigma &= \pi \left(\frac{16}{495} \tau^2 + \frac{32}{1155} J \right), \\
\int Y^2 |M|^2 d\sigma &= \pi \left(\frac{8}{315} \tau^2 + \frac{16}{105} J \right), \\
\int |v|^4 d\sigma &= \pi \left(\frac{16}{315} \tau^2 + \frac{32}{315} J \right), \\
\int |v|^2 |M|^2 d\sigma &= \pi \left(\frac{8}{105} \tau^2 + \frac{32}{105} J \right), \\
\int |M|^4 d\sigma &= \pi \left(\frac{4}{15} \tau^2 + \frac{8}{15} J \right).
\end{aligned} \tag{7.3}$$

8. Assembly of the quartic identity

From (4.7),

$$q = 64a + 144b - 72c.$$

Therefore

$$q^2 = 4096a^2 + 18432ab - 9216ac + 20736b^2 - 20736bc + 5184c^2. \tag{8.1}$$

Substitute the six formulas in (7.3). Dividing first by π , the coefficient of τ^2 is

$$\begin{aligned}
&4096 \frac{16}{715} + 18432 \frac{16}{495} - 9216 \frac{8}{315} \\
&+ 20736 \frac{16}{315} - 20736 \frac{8}{105} + 5184 \frac{4}{15} = \frac{1310464}{1001},
\end{aligned} \tag{8.2}$$

while the coefficient of J is

$$\begin{aligned}
&4096 \frac{96}{5005} + 18432 \frac{32}{1155} - 9216 \frac{16}{105} \\
&+ 20736 \frac{32}{315} - 20736 \frac{32}{105} + 5184 \frac{8}{15} = -\frac{2265600}{1001}.
\end{aligned} \tag{8.3}$$

Hence

$$\int_{S^2} q^2 d\sigma = \frac{\pi}{1001} (1310464\tau^2 - 2265600J). \tag{8.4}$$

Since

$$J = |B|^2 = |B_0|^2 + \frac{1}{3}\tau^2, \tag{8.5}$$

we obtain

$$\begin{aligned}\int_{S^2} q^2 d\sigma &= \frac{\pi}{1001} \left(\left(1310464 - \frac{2265600}{3} \right) \tau^2 - 2265600 |B_0|^2 \right) \\ &= \frac{\pi}{1001} (555264 \tau^2 - 2265600 |B_0|^2).\end{aligned}\tag{8.6}$$

This proves (1.6).

9. The determinant estimate and equality case

Because $|B_0|^2 \geq 0$, (8.6) gives

$$\|q\|_2^2 \leq \frac{555264\pi}{1001} \tau^2.\tag{9.1}$$

From (1.5),

$$\|C\|_2^4 = \left(\frac{176\pi}{7} \right)^2 \tau^2.\tag{9.2}$$

Eliminating τ^2 between (9.1) and (9.2),

$$\begin{aligned}\|4 \det C\|_2^2 &\leq \frac{555264\pi/1001}{(176\pi/7)^2} \|C\|_2^4 \\ &= \frac{15183}{17303\pi} \|C\|_2^4.\end{aligned}\tag{9.3}$$

Thus (1.7) is proved.

Equality in (9.3) holds exactly when $B_0=0$. This condition is nonempty. For example, take $Y(x,y,z)=xyz$. In the normalization (1.2), the six components $T_{\{123\}}$ and their permutations equal $1/6$; all other components vanish. Then

$$\tau = \frac{1}{6}, \quad B = \frac{1}{18} I = \frac{\tau}{3} I,$$

so $B_0=0$.

10. Verification map

The hand derivation corresponds to the code as follows.

mathematical step	implementation
general symmetric trace-free tensor and harmonicity	<code>python/verify_degree3_identity.py</code>
projected formula $C=6PMP-2YP$	both exact identity scripts
compact formula (4.7)	<code>python/verify_degree3_identity_alt.py</code> and numerical tests
exact sphere moments and identities (1.5)-(1.6)	<code>python/verify_degree3_identity.py</code>
pairing counts in Section 7	<code>python/verify_pairing_counts.py</code> ; matching R script
determinant ratio and final simplification	<code>python/verify_bound_arithmetic.py</code>

The exact scripts and this derivation use the same conventions:

- $Y=T[u,u,u]$, with no hidden factorial;
- C is represented by a 3×3 extension that vanishes on u ;
- $q=4 \det_{\{u^\perp\}} C$;
- $|C|^2=\text{tr}(C^2)$;
- $\text{sigma}(S^2)=4\pi$.

11. Remaining review status

This file closes the principal internal write-up gap identified as T1-T4 in `proof/PROOF_LEDGER.md`. It has been checked against two exact symbolic implementations and independent Python/R numerical tests, incorporated into `paper/degree3_spectral_refinement.tex`, and reviewed in `proof/PROOF_AUDIT.md`. It has not received outside expert review. It should therefore be described as a **formal derivation in an AI-assisted research draft**, not as a peer-reviewed or independently certified theorem.